
Project 03

LINUX PROJECT REPORT



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TABLE OF CONTENTS

1. Introduction	3
1.1 Project Overview	3
2. Detailed Design	5
2.1 Flowchart of System Architecture	5
2.2 IP Addressing Scheme	5
3. DNS Implementation	6
3.1 Introduction to DNS	6
3.2 Behavior of DNS Protocol	6
3.3 Commands Used	6
3.4 Integration and Testing Results	7
4. DHCP Implementation	15
4.1 Introduction to DHCP	15
4.2 Behavior of DHCP Protocol	15
4.3 Commands Used	15
4.4 Integration and Testing Results	16
5. Web Server Implementation	20
5.1 Introduction to Web Server	20
5.2 Behavior of Web Server Protocol	20
5.3 Commands Used	20
5.4 Integration and Testing Results	21
6. Firewall Implementation	22
6.1 Introduction to Firewall	22
6.2 Behavior of Firewall	22
6.3 Commands Used	22
6.4 Integration and Testing Results	23
7. Backup Implementation	24
7.1 Introduction to Backup System	24
7.2 Behavior of Backup Protocol	24
7.3 Commands Used	24
7.4 Integration and Testing Results	25
8. MITM Implementation	26
8.1 Introduction	26
8.2 Commands Used	26
8.3 Test Results	26
9. IPSec VPN Tunneling	27
9.1 Introduction	27
9.2 Commands Used	27
9.3 Test Results	28
10. Linux System Topics	29
10.1 File System (tar, gunzip)	29
10.2 Memory Management	29
10.3 Processes	29
11. Future Scope	30
12. References	31

1. INTRODUCTION

1.1. PROJECT OVERVIEW

This comprehensive Linux networking infrastructure project involves designing, implementing, and documenting a complete network solution for a start-up company based in Boston. The project demonstrates advanced skills in network administration, server configuration, and security implementation using Linux-based systems.

Core Components

DNS Implementation

The project establishes a robust Domain Name System using Bind9, featuring both Master and Slave DNS servers with automatic synchronization. The implementation includes both forward and reverse lookup zones supporting both IPv4 and IPv6 addressing schemes, ensuring reliable name resolution services across the network.

DHCP Services

A Dynamic Host Configuration Protocol server provides automated IP address management with sophisticated features including address pools, exclusion ranges, and permanent reservations for critical infrastructure devices. The DHCP implementation supports both IPv4 and IPv6 addressing with configurable lease times tailored to the organization's needs.

Web Server & Firewall

The project incorporates a web server deployment secured by a properly configured firewall. Security measures implement best practices in network protection including IP filtering, protocol filtering, and access controls - all accomplished using native Linux command-line tools and configuration. This demonstrates how to achieve robust security even with limited resources.

Backup System

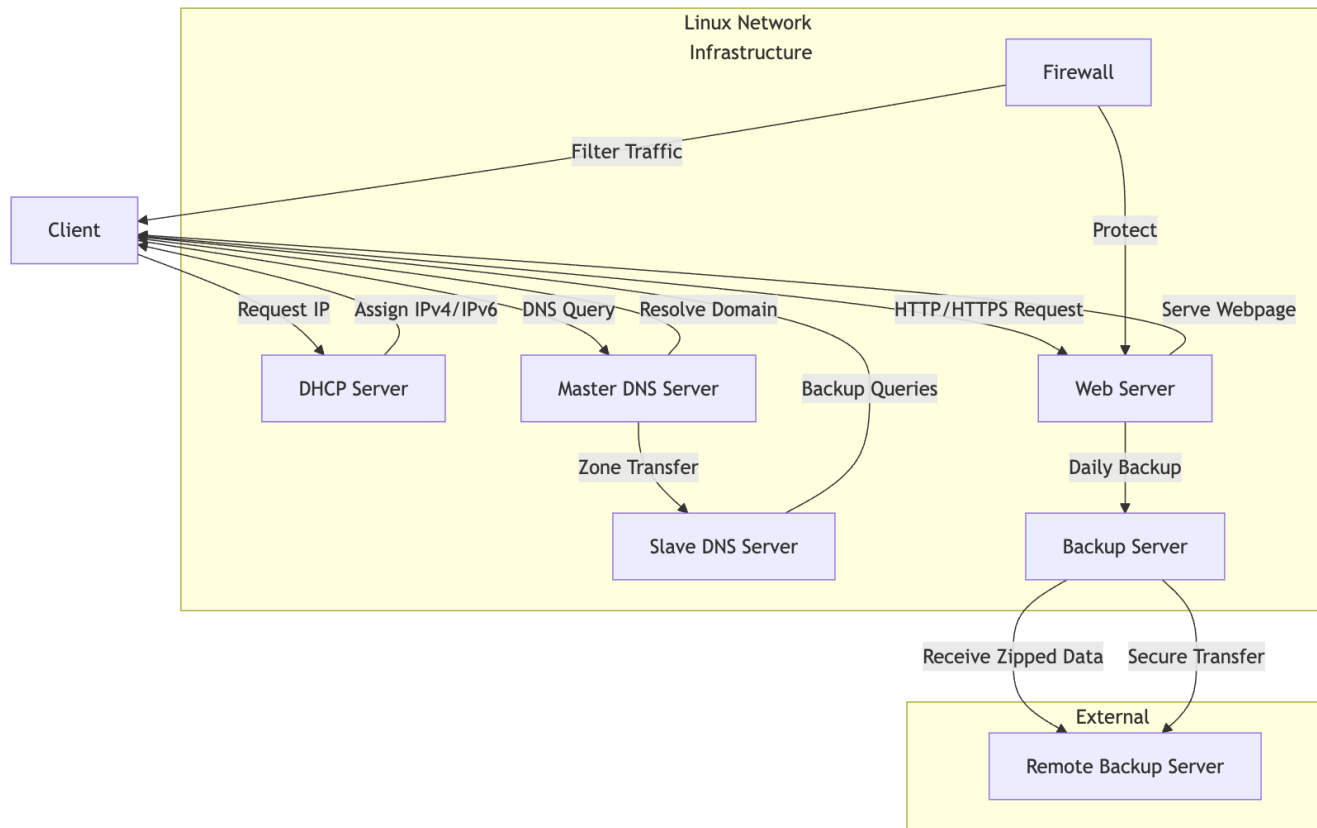
A systematic backup solution ensures business continuity through automated, scheduled backups with compression and secure transfer to a separate backup server. This implementation showcases Linux's powerful scripting and scheduling capabilities to protect critical business data.

The project integrates these components to demonstrate a cohesive system where the DHCP server assigns IP addresses, the DNS resolves domain queries, the web server delivers content, the firewall protects the network, and the backup system ensures data reliability. A comprehensive test plan, including screenshots and examples, validates the functionality of each module and their interoperability.

Additionally, the project explores Linux system concepts such as file systems (tar, gunzip), memory management, and processes, providing a holistic understanding of the underlying platform. The report includes a detailed design with flowcharts, an IP addressing scheme, and a discussion on future scalability, ensuring the system's relevance for the startup's growth.

This project showcases practical telecommunications engineering skills, emphasizing open-source tools, automation, and security within budget constraints, preparing the team for real-world network deployment challenges.

FLOW DIAGRAM:



2. DETAILED DESIGN

2.1. FLOWCHART OF SYSTEM ARCHITECTURE

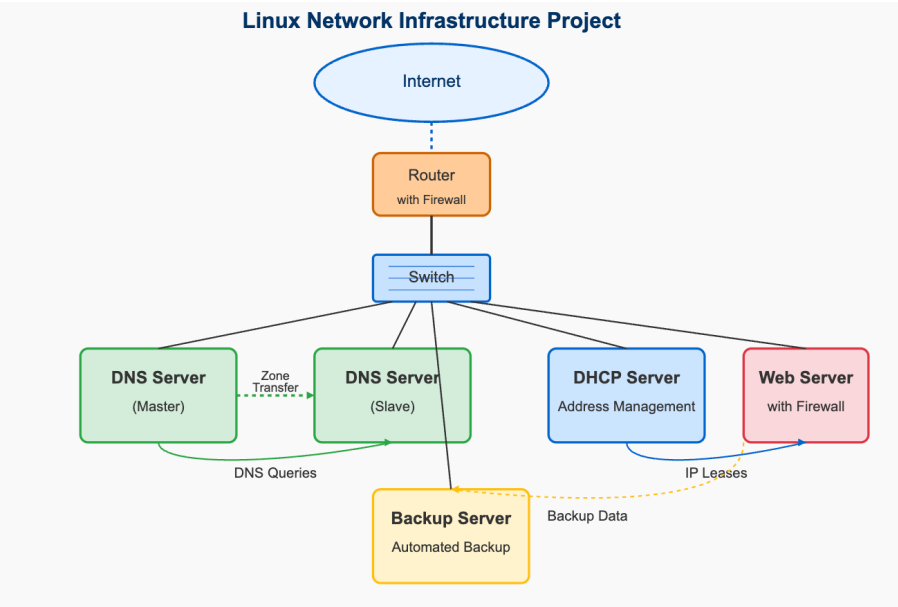


Fig: Block diagram of the network infrastructure

2.2. IP ADDRESSING SCHEME

Server Configuration			
Service	IP Address	IPv6 Address	Function
DNS Master	192.168.27.130	2001:db8:27::130	Primary DNS server for domain resolution
DNS Slave	192.168.27.131	2001:db8:27::131	Secondary DNS with automatic updates
DHCP Server	192.168.27.129	2001:db8:27::129	Dynamic IP address allocation
Web Server	192.168.27.132	2001:db8:27::132	Apache web hosting with firewall
Backup Server	192.168.27.134	2001:db8:27::134	Daily backup of all server configurations

Client Devices			
Client	IP Address	IPv6 Address	Status
Client 1	192.168.27.133	2001:db8:27::170	Connected
Client 2	192.168.27.135	2001:db8:27::1b8	Connected

Fig: IP addressing scheme

3. DNS IMPLEMENTATION

3.1. INTRODUCTION TO DNS

Domain Name System (DNS) serves as the internet's phonebook, translating human-readable domain names like "group1-startup.com" into machine-understandable IP addresses that computers use to identify each other. This distributed hierarchical system operates through a global network of DNS servers that work together to resolve queries, starting from root servers at the top of the hierarchy down to authoritative nameservers that hold specific domain information. DNS is fundamental to network functionality as it enables users to access websites and services using memorable names instead of numerical addresses, while supporting additional record types that facilitate email routing, service discovery, and other critical internet functions. In a corporate environment like the Group1 start-up in this project, implementing a properly configured DNS server is essential for internal network operations, allowing employees to access resources by name rather than remembering complex IP addresses.

3.2. BEHAVIOR OF DNS

- **Recursive Query:** In a recursive DNS query, the client requests the DNS resolver to obtain the final answer. If the resolver does not already have the requested information cached, it continues to query other DNS servers on behalf of the client until it retrieves the correct IP address and returns it.
- **Iterative Query:** In an iterative query, the DNS resolver responds with a referral to another DNS server that is closer to the authoritative source. The client then sends a new query to the referred server. This process continues until the client eventually receives the required IP address.
- **Caching and Optimization:** DNS resolvers cache responses from previous queries to improve efficiency and reduce network traffic. The duration for which a response is cached depends on the time-to-live (TTL) value specified in the DNS records. For most standard queries, DNS uses the User Datagram Protocol (UDP) over port 53 due to its low overhead and speed. However, in cases where reliability is critical or the response size exceeds the limit of UDP, Transmission Control Protocol (TCP) is used instead. This dual-protocol mechanism helps maintain both performance and reliability across various use cases.

3.3. COMMANDS USED

DNS Master Server Configuration (192.168.27.130)

Step Description	Command
Install BIND9 packages	<code>sudo apt install bind9 bind9utils bind9-doc</code>
Configure network settings	<code>sudo nano /etc/netplan/01-netcfg.yaml</code>
Apply network settings	<code>sudo netplan apply</code>
Configure BIND9 main options	<code>sudo nano /etc/bind/named.conf.options</code>
Configure local zones	<code>sudo nano /etc/bind/named.conf.local</code>
Create forward zone file	<code>sudo mkdir -p /var/lib/bind</code> <code>sudo nano /var/lib/bind/db.group1-startup.com</code>
Create IPv4 reverse zone file	<code>sudo nano /var/lib/bind/db.192.168.27</code>
Create IPv6 reverse zone file	<code>sudo nano /var/lib/bind/db.2001.db8.27</code>
Set permissions for zone files	<code>sudo chown -R bind:bind /var/lib/bind</code>

	<code>sudo chmod -R 755 /var/lib/bind</code>
Check main configuration	<code>sudo named-checkconf</code>
Check forward zone file	<code>sudo named-checkzone group1-startup.com /var/lib/bind/db.group1-startup.com</code>
Check IPv4 reverse zone	<code>sudo named-checkzone 27.168.192.in-addr.arpa /var/lib/bind/db.192.168.27</code>
Check IPv6 reverse zone	<code>sudo named-checkzone 0.0.0.0.0.0.0.0.7.2.0.0.8.b.d.0.1.0.0.2.ip6.arpa /var/lib/bind/db.2001.db8.27</code>
Start and enable BIND9 service	<code>sudo systemctl restart bind9</code> <code>sudo systemctl enable bind9</code> <code>sudo systemctl status bind9</code>

DNS Slave Server Configuration (192.168.27.131)

Step Description	Command
Install BIND9 packages	<code>sudo apt install bind9 bind9utils bind9-doc</code>
Configure network settings	<code>sudo nano /etc/netplan/01-netcfg.yaml</code>
Apply network settings	<code>sudo netplan apply</code>
Configure BIND9 main options	<code>sudo nano /etc/bind/named.conf.options</code>
Configure local zones	<code>sudo nano /etc/bind/named.conf.local</code>
Start and enable BIND9 service	<code>sudo systemctl restart bind9</code> <code>sudo systemctl enable bind9</code> <code>sudo systemctl status bind9</code>
Testing DNS Resolution	<code>dig @192.168.27.130 ns1.group1-startup.com</code> <code>dig @192.168.27.130 www.group1-startup.com</code> <code>dig @192.168.27.130 -x 192.168.27.130</code> <code>dig @192.168.27.131 ns1.group1-startup.com</code> <code>dig @192.168.27.131 www.group1-startup.com</code> <code>dig @192.168.27.131 -x 192.168.27.131</code>

3.4. TEST RESULTS

```

group1@ubuntu: /etc/netplan
File Edit View Search Terminal Help
GNU nano 2.9.3                                01-network-manager-all.yaml

# Let NetworkManager manage all devices on this system
network:
  version: 2
  renderer: NetworkManager
  ethernets:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.27.130/24
        - 2001:db8:27::130/64
      gateway4: 192.168.27.2
      gateway6: 2001:db8:27::2
      nameservers:
        addresses: [127.0.0.1, 8.8.8.8, 8.8.4.4, "2001:4860:4860::8888", "2001:4860:4860::8844", 192.168.27.131, "2001:db8:27::131"]

```

Fig: DNS Master Network Configuration

```

group1@ubuntu: ~
File Edit View Search Terminal Help
group1@ubuntu:~$ sudo systemctl status bind9
● bind9.service - BIND Domain Name Server
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2025-04-13 16:13:50 PDT; 1 day 4h ago
     Docs: man:named(8)
  Main PID: 9174 (named)
    Tasks: 5 (limit: 2281)
   CGroup: /system.slice/bind9.service
           └─9174 /usr/sbin/named -f -u bind

Apr 14 17:55:25 ubuntu named[9174]: configuring command channel from '/etc/bind/rndc.key'
Apr 14 17:55:25 ubuntu named[9174]: configuring command channel from '/etc/bind/rndc.key'
Apr 14 17:55:25 ubuntu named[9174]: reloading configuration succeeded
Apr 14 17:55:25 ubuntu named[9174]: scheduled loading new zones
Apr 14 17:55:25 ubuntu named[9174]: any newly configured zones are now loaded
Apr 14 17:55:25 ubuntu named[9174]: running
Apr 14 17:55:25 ubuntu named[9174]: managed-keys-zone: Key 20326 for zone . acceptance timer complete: key now trusted
Apr 14 17:55:25 ubuntu named[9174]: managed-keys-zone: Key 38696 for zone . acceptance timer complete: key now trusted
Apr 14 17:58:13 ubuntu named[9174]: client @0x7f11240aa180 127.0.0.1#46662 (connectivity-check.ubuntu.com): query (cache) 'connectivity-check.ubuntu.com/AAAA/IN' denied
Apr 14 17:58:15 ubuntu named[9174]: success resolving 'mmx-ds.cdn.whatsapp.net/A' (in 'net?') after disabling EDNS
group1@ubuntu:~$

```

Fig: Bind9 DNS Server Status

```

group1@ubuntu: ~
File Edit View Search Terminal Help
GNU nano 2.9.3 /etc/bind/named.conf.options

#options {
#   directory "/var/cache/bind";

    // Listen on all interfaces for both IPv4 and IPv6
    listen-on { any; };
    listen-on-v6 { any; };

    // Allow queries from your local network
    allow-query { 192.168.27.0/24; 2001:db8:27::/64; };

    // Allow recursive queries only from local network
    recursion yes;
    allow-recursion { 192.168.27.0/24; 2001:db8:27::/64; };

    // Forward queries we can't resolve to Google DNS
    forwarders {
        8.8.8.8;
        8.8.4.4;
        192.168.27.130;
        192.168.27.131;
    };

    // Configure slave DNS server to get zone transfers
    allow-transfer { 192.168.27.131; 2001:db8:27::131; };

    dnssec-validation auto;
    auth-nxdomain no;

    // Optional: increase security
    version none;
    hostname none;
    server-id none;
};

```

Fig: DNS Master Configuration (Options)

```

group1@ubuntu: ~
File Edit View Search Terminal Help
GNU nano 2.9.3 /etc/bind/named.conf.local

//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
//include "/etc/bind/zones.rfc1918";

// Forward zone for boston-startup.com
zone "group1-startup.com" {
    type master;
    file "/var/lib/bind/db.group1-startup.com";
    allow-transfer { 192.168.27.131; 2001:db8:27::131; };
    also-notify { 192.168.27.131; 2001:db8:27::131; };
};

// Reverse zone for IPv4 addresses
zone "27.168.192.in-addr.arpa" {
    type master;
    file "/var/lib/bind/db.192.168.27";
    allow-transfer { 192.168.27.131; 2001:db8:27::131; };
    also-notify { 192.168.27.131; 2001:db8:27::131; };
};

// Reverse zone for IPv6 addresses
zone "0.0.0.0.0.0.0.7.2.0.8.b.d.0.1.0.0.2.ip6.arpa" {
    type master;
    file "/var/lib/bind/db.2001.db8.27";
    allow-transfer { 192.168.27.131; 2001:db8:27::131; };
    also-notify { 192.168.27.131; 2001:db8:27::131; };
};

```

Fig: DNS Master Configuration (Local)

```

group1@ubuntu: ~
File Edit View Search Terminal Help
GNU nano 2.9.3 /var/lib/bind/db.192.168.27

$TTL      604800
@         IN      SOA      ns1.group1-startup.com. admin.group1-startup.com. (
                                2025041201      ; Serial
                                604800           ; Refresh
                                86400            ; Retry
                                2419200          ; Expire
                                604800 )         ; Negative Cache TTL

; Name servers
@         IN      NS       ns1.group1-startup.com.
@         IN      NS       ns2.group1-startup.com.

; PTR Records
130       IN      PTR      ns1.group1-startup.com.
131       IN      PTR      ns2.group1-startup.com.
129       IN      PTR      dhcp.group1-startup.com.
132       IN      PTR      www.group1-startup.com.
133       IN      PTR      client1.group1-startup.com.
135       IN      PTR      client2.group1-startup.com.
134       IN      PTR      backup.group1-startup.com.

```

Fig: Bind9 Database

```

group1@ubuntu: ~
File Edit View Search Terminal Help
GNU nano 2.9.3 /var/lib/bind/db.2001.db8.27

$TTL      604800
@         IN      SOA      ns1.group1-startup.com. admin.group1-startup.com. (
                                2025041201      ; Serial
                                604800          ; Refresh
                                86400           ; Retry
                                2419200        ; Expire
                                604800 )       ; Negative Cache TTL

; Name servers
@         IN      NS       ns1.group1-startup.com.
@         IN      NS       ns2.group1-startup.com.

; PTR Records for IPv6 addresses (in nibble format)
0.3.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0 IN PTR ns1.group1-startup.com.
1.3.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0 IN PTR ns2.group1-startup.com.
9.2.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0 IN PTR dhcp.group1-startup.com.
2.3.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0 IN PTR www.group1-startup.com.
0.7.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0 IN PTR client1.group1-startup.com.
3.f.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0 IN PTR client2.group1-startup.com.

4.3.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0 IN PTR backup.group1-startup.com.

```

```

group1@ubuntu: ~
File Edit View Search Terminal Help
GNU nano 2.9.3 /var/lib/bind/db.group1-startup.com

STTL      604800
@         IN      SOA      ns1.group1-startup.com. admin.group1-startup.com. (
                        2025041201      ; Serial (use today's date)
                        604800          ; Refresh
                        86400           ; Retry
                        2419200         ; Expire
                        604800 )        ; Negative Cache TTL

; Name servers
@         IN      NS       ns1.group1-startup.com.
@         IN      NS       ns2.group1-startup.com.

; A records for name servers
ns1       IN      A        192.168.27.130
ns1       IN      AAAA     2001:db8:27::130
ns2       IN      A        192.168.27.131
ns2       IN      AAAA     2001:db8:27::131

; Other A records
@         IN      A        192.168.27.130
@         IN      AAAA     2001:db8:27::130
www       IN      A        192.168.27.132
www       IN      AAAA     2001:db8:27::132
dhcp      IN      A        192.168.27.129
dhcp      IN      AAAA     2001:db8:27::129
client1   IN      A        192.168.27.133
client1   IN      AAAA     2001:db8:27::170
client2   IN      A        192.168.27.135
client2   IN      AAAA     2001:db8:27::1b8
backup    IN      A        192.168.27.134
backup    IN      AAAA     2001:db8:27::134

```

```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
group1@ubuntu:~$ sudo named-checkzone group1-startup.com /var/lib/bind/db.group1-startup.com  
zone group1-startup.com/IN: loaded serial 2025041201  
OK  
group1@ubuntu:~$ sudo named-checkzone 27.168.192.in-addr.arpa /var/lib/bind/db.192.168.27  
zone 27.168.192.in-addr.arpa/IN: loaded serial 2025041201  
OK  
group1@ubuntu:~$ sudo named-checkzone 0.0.0.0.0.0.0.7.2.0.0.8.b.d.0.1.0.0.2.ip6.arpa /var/lib/bind/db.2001.db8.27  
zone 0.0.0.0.0.0.0.7.2.0.0.8.b.d.0.1.0.0.2.ip6.arpa/IN: loaded serial 2025041201  
OK  
group1@ubuntu:~$ █
```

Fig: Checkzones

```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
group1@ubuntu:~$  
group1@ubuntu:~$  
group1@ubuntu:~$ nslookup client1.group1-startup.com  
Server: 127.0.0.53  
Address: 127.0.0.53#53  
  
Non-authoritative answer:  
Name: client1.group1-startup.com  
Address: 192.168.27.133  
Name: client1.group1-startup.com  
Address: 2001:db8:27::170  
  
group1@ubuntu:~$ nslookup client2.group1-startup.com  
Server: 127.0.0.53  
Address: 127.0.0.53#53  
  
Non-authoritative answer:  
Name: client2.group1-startup.com  
Address: 192.168.27.135  
Name: client2.group1-startup.com  
Address: 2001:db8:27::1b8  
  
group1@ubuntu:~$ nslookup client2.group1-startup.com 192.168.27.130  
Server: 192.168.27.130  
Address: 192.168.27.130#53  
  
Name: client2.group1-startup.com  
Address: 192.168.27.135  
Name: client2.group1-startup.com  
Address: 2001:db8:27::1b8  
  
group1@ubuntu:~$ nslookup client2.group1-startup.com 192.168.27.130  
Server: 192.168.27.130  
Address: 192.168.27.130#53  
  
Name: client2.group1-startup.com  
Address: 192.168.27.135  
Name: client2.group1-startup.com  
Address: 2001:db8:27::1b8  
  
group1@ubuntu:~$ nslookup client1.group1-startup.com 192.168.27.130  
Server: 192.168.27.130  
Address: 192.168.27.130#53  
  
Name: client1.group1-startup.com  
Address: 192.168.27.133  
Name: client1.group1-startup.com  
Address: 2001:db8:27::170
```

Fig: nslookup from client machines

```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
GNU nano 2.9.3 /etc/netplan/01-network-manager-all.yaml  
# Let NetworkManager manage all devices on this system  
#network:  
# version: 2  
# renderer: NetworkManager  
  
network:  
  version: 2  
  renderer: NetworkManager  
  ethernet:  
    ens33:  
      dhcp4: no  
      addresses:  
        - 192.168.27.131/24  
        - 2001:db8:27::131/64  
      gateway4: 192.168.27.2  
      gateway6: 2001:db8:27::2  
      nameservers:  
        addresses: [127.0.0.1, 8.8.8.8, 8.8.4.4, 192.168.27.130, "2001:4860:4860::8888", "2001:4860:4860::8844", "2001:db8:27::130"]
```

Fig: DNS Slave Server Network Configuration

```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
group1@ubuntu:~$ sudo systemctl status bind9  
● bind9.service - BIND Domain Name Server  
   Loaded: loaded (/lib/systemd/system/bind9.service; enabled; vendor preset: enabled)  
   Active: active (running) since Sat 2025-04-12 10:45:58 PDT; 2 days ago  
     Docs: man:named(8)  
    Main PID: 12473 (named)  
      Tasks: 5 (limit: 2284)  
    CGroup: /system.slice/bind9.service  
            └─12473 /usr/sbin/named -f -u bind  
  
Apr 14 17:57:55 ubuntu named[12473]: resolver priming query complete  
Apr 14 17:58:10 ubuntu named[12473]: resolver priming query complete  
Apr 14 17:58:14 ubuntu named[12473]: success resolving 'mmx-ds.cdn.whatsapp.net/A' (in 'net?') after disabling EDNS  
Apr 14 17:58:14 ubuntu named[12473]: resolver priming query complete  
Apr 14 17:58:14 ubuntu named[12473]: checkhints: b.root-servers.net/A (170.247.170.2) missing from hints  
Apr 14 17:58:14 ubuntu named[12473]: checkhints: b.root-servers.net/A (192.228.79.201) extra record in hints  
Apr 14 17:58:14 ubuntu named[12473]: checkhints: b.root-servers.net/AAAA (2801:1b8:10::b) missing from hints  
Apr 14 17:58:14 ubuntu named[12473]: checkhints: b.root-servers.net/AAAA (2001:500:84::b) extra record in hints  
Apr 14 17:58:14 ubuntu named[12473]: checkhints: l.root-servers.net/AAAA (2001:500:9f::42) missing from hints  
Apr 14 17:58:14 ubuntu named[12473]: checkhints: l.root-servers.net/AAAA (2001:500:3::42) extra record in hints  
group1@ubuntu:~$
```

Fig: Bind9 DNS Slave Server Status

```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
GNU nano 2.9.3 /etc/bind/named.conf.options  
# listen-on-v6 { any; };  
  
options {  
  directory "/var/cache/bind";  
  
  listen-on { any; };  
  listen-on-v6 { any; };  
  
  allow-query { 192.168.27.0/24; 2001:db8:27::/64; };  
  
  recursion yes;  
  allow-recursion { 192.168.27.0/24; 2001:db8:27::/64; };  
  
  // Forward DNS queries to Master DNS server when not authoritative  
  forwarders {  
    192.168.27.130;  
    8.8.8.8;  
  };  
  
  dnssec-validation auto;  
  auth-nxdomain no;  
};
```

Fig: DNS Slave Configuration (Options)

```

group1@ubuntu: ~
File Edit View Search Terminal Help
GNU nano 2.9.3 /etc/bind/named.conf.local

//
/// Do any local configuration here
///
/// Consider adding the 1918 zones here, if they are not used in your
/// organization
///include "/etc/bind/zones.rfc1918";

// Forward zone for group1-startup.com
zone "group1-startup.com" {
    type slave;
    file "db.group1-startup.com";
    masters { 192.168.27.130; };
};

// Reverse zone for IPv4 addresses
zone "27.168.192.in-addr.arpa" {
    type slave;
    file "db.192.168.27";
    masters { 192.168.27.130; };
};

// Reverse zone for IPv6 addresses
zone "0.0.0.0.0.0.0.0.7.2.0.0.8.b.d.0.1.0.0.2.ip6.arpa" {
    type slave;
    file "db.2001.db8.27";
    masters { 192.168.27.130; };
};

```

Fig: DNS Slave Configuration (Local)

```

group1@ubuntu:~$ dig @192.168.27.131 ns1.group1-startup.com

; <<>> DiG 9.11.3-1ubuntu1.18-Ubuntu <<>> @192.168.27.131 ns1.group1-startup.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 57736
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 2, ADDITIONAL: 4

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
; COOKIE: 19f0068d62dffce929bc126467fddcf1ab0714bb10e28c53 (good)
;; QUESTION SECTION:
ns1.group1-startup.com.                IN      A

;; ANSWER SECTION:
ns1.group1-startup.com. 604800  IN      A      192.168.27.130

;; AUTHORITY SECTION:
group1-startup.com.    604800  IN      NS      ns1.group1-startup.com.
group1-startup.com.    604800  IN      NS      ns2.group1-startup.com.

;; ADDITIONAL SECTION:
ns2.group1-startup.com. 604800  IN      A      192.168.27.131
ns1.group1-startup.com. 604800  IN      AAAA    2001:db8:27::130
ns2.group1-startup.com. 604800  IN      AAAA    2001:db8:27::131

;; Query time: 9 msec
;; SERVER: 192.168.27.131#53(192.168.27.131)
;; WHEN: Mon Apr 14 21:13:37 PDT 2025
;; MSG SIZE rcvd: 199

```

Fig: DNS Test Results

```

group1@ubuntu:~$ dig @192.168.27.131 www.group1-startup.com

; <<>> DiG 9.11.3-1ubuntu1.18-Ubuntu <<>> @192.168.27.131 www.group1-startup.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 3627
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 2, ADDITIONAL: 5

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:: udp: 4096
; COOKIE: bf7c3a99941dc8f2c2aeea2167fddd04ea4a466b9b60b673 (good)
;; QUESTION SECTION:
;www.group1-startup.com.                IN      A

;; ANSWER SECTION:
www.group1-startup.com. 604800  IN      A      192.168.27.132

;; AUTHORITY SECTION:
group1-startup.com.    604800  IN      NS      ns2.group1-startup.com.
group1-startup.com.    604800  IN      NS      ns1.group1-startup.com.

;; ADDITIONAL SECTION:
ns1.group1-startup.com. 604800  IN      A      192.168.27.130
ns2.group1-startup.com. 604800  IN      A      192.168.27.131
ns1.group1-startup.com. 604800  IN      AAAA    2001:db8:27::130
ns2.group1-startup.com. 604800  IN      AAAA    2001:db8:27::131

;; Query time: 1 msec
;; SERVER: 192.168.27.131#53(192.168.27.131)
;; WHEN: Mon Apr 14 21:13:56 PDT 2025
;; MSG SIZE rcvd: 219

```

Fig: DNS Test Results

```

group1@ubuntu:~$ dig @192.168.27.131 -x 192.168.27.131

; <<>> DiG 9.11.3-1ubuntu1.18-Ubuntu <<>> @192.168.27.131 -x 192.168.27.131
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 22124
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 2, ADDITIONAL: 5

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:: udp: 4096
; COOKIE: afc14418f2b92d03f07c4b3d67fddd1574ac3f5b8d79fd6e (good)
;; QUESTION SECTION:
;131.27.168.192.in-addr.arpa.          IN      PTR

;; ANSWER SECTION:
131.27.168.192.in-addr.arpa. 604800  IN      PTR      ns2.group1-startup.com.

;; AUTHORITY SECTION:
27.168.192.in-addr.arpa. 604800  IN      NS      ns2.group1-startup.com.
27.168.192.in-addr.arpa. 604800  IN      NS      ns1.group1-startup.com.

;; ADDITIONAL SECTION:
ns1.group1-startup.com. 604800  IN      A      192.168.27.130
ns2.group1-startup.com. 604800  IN      A      192.168.27.131
ns1.group1-startup.com. 604800  IN      AAAA    2001:db8:27::130
ns2.group1-startup.com. 604800  IN      AAAA    2001:db8:27::131

;; Query time: 3 msec
;; SERVER: 192.168.27.131#53(192.168.27.131)
;; WHEN: Mon Apr 14 21:14:13 PDT 2025
;; MSG SIZE rcvd: 240

```

Fig: DNS Test Results

4. DHCP IMPLEMENTATION

4.1. INTRODUCTION TO DHCP

The Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to automatically assign IP addresses and other network configuration parameters to devices on a network, allowing them to communicate efficiently with other IP networks. By automating the assignment process, DHCP eliminates the need for manual IP configuration by network administrators, reducing errors and simplifying device connectivity. It operates based on a client-server model where the DHCP server dynamically allocates IP addresses from a defined range (or pool) to client devices for a specified lease time. Along with the IP address, it can also provide essential configuration details such as subnet mask, default gateway, and DNS server addresses.

4.2. BEHAVIOR OF DHCP

DHCP Discover:

- The client device broadcasts a **DHCP Discover** message to locate available DHCP servers.
- This message is sent to the network using the IP address 255.255.255.255 since the client doesn't yet have an IP.

DHCP Offer:

- DHCP servers on the network respond with a **DHCP Offer**.
- This message contains an available IP address and other configuration details like subnet mask, gateway, and DNS servers.

DHCP Request:

- The client replies with a **DHCP Request** message, indicating its acceptance of one of the offers.
- This message is also broadcasted to inform all DHCP servers that it has chosen a particular server's offer.

DHCP Acknowledgment:

- The selected DHCP server sends a **DHCP Acknowledgment (ACK)** message.
- This confirms the assignment, and the client can now use the provided IP address and configurations.

4.3. COMMANDS USED

Step Description	Commands
Install DHCP server	<code>sudo apt install isc-dhcp-server</code>
Edit network configuration	<code>sudo nano /etc/netplan/01-network-manager-all.yaml</code>
Apply network configuration	<code>sudo netplan apply</code>
Edit DHCP configuration	<code>sudo nano /etc/dhcp/dhcpd.conf</code>
Edit DHCPv6 configuration	<code>sudo nano /etc/dhcp/dhcpd6.conf</code>
Edit DHCP server settings	<code>sudo nano /etc/default/isc-dhcp-server</code>
Check configuration syntax	<code>sudo dhcpd -t</code>
Check DHCPv6 configuration syntax	<code>sudo dhcpd -6 -t -cf /etc/dhcp/dhcpd6.conf</code>
Create DHCPv6 leases file	<code>sudo touch /var/lib/dhcp/dhcpd6.leases</code>
Set permissions for leases	<code>sudo chmod 664 /var/lib/dhcp/dhcpd6.leases</code>
Start/restart DHCP service	<code>sudo systemctl restart isc-dhcp-server</code>
Start/restart DHCPv6 service	<code>sudo systemctl restart isc-dhcp-server6</code>
Enable DHCP service	<code>sudo systemctl enable isc-dhcp-server</code>
Enable DHCPv6 service	<code>sudo systemctl enable isc-dhcp-server6</code>

Check service status	<code>sudo systemctl status isc-dhcp-server</code>
Check DHCPv6 service status	<code>sudo systemctl status isc-dhcp-server6</code>
View service logs	<code>sudo journalctl -u isc-dhcp-server</code>
View DHCPv6 service logs	<code>sudo journalctl -u isc-dhcp-server6</code>
View DHCP leases	<code>sudo cat /var/lib/dhcp/dhcpd.leases</code>
View DHCPv6 leases	<code>sudo cat /var/lib/dhcp/dhcpd6.leases</code>
Release and renew IP (client)	<code>sudo dhclient -r</code> <code>sudo dhclient</code>
Release and renew IPv6 (client)	<code>sudo dhclient -6 -r</code> <code>sudo dhclient -6</code>
View network interfaces	<code>ifconfig or ip addr</code>
Flush IPv6 addresses (client)	<code>sudo ip -6 addr flush dev ens33</code>

4.4. TEST RESULTS

```
group1@ubuntu: /etc/netplan
File Edit View Search Terminal Help
GNU nano 2.9.3                                01-network-manager-all.yaml

# Let NetworkManager manage all devices on this system
network:
  version: 2
  renderer: NetworkManager
  ethernet:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.27.129/24
        - 2001:db8:27::129/64
      gateway4: 192.168.27.2
      gateway6: 2001:db8:27::2
      nameservers:
        addresses: [8.8.8.8, 8.8.4.4, "2001:4860:4860::8888", "2001:4860:4860::8844", 192.168.27.130, 192.168.27.131, "2001:db8:27::130", "2001:db8:27::131"]
```

Fig: DHCP Network Configuration

```
group1@ubuntu:~$ sudo systemctl status isc-dhcp-server6
● isc-dhcp-server6.service - ISC DHCP IPv6 server
   Loaded: loaded (/lib/systemd/system/isc-dhcp-server6.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2025-04-11 22:23:51 PDT; 3 days ago
     Docs: man:dhcpd(8)
    Main PID: 6213 (dhcpd)
      Tasks: 1 (limit: 2281)
   CGroup: /system.slice/isc-dhcp-server6.service
           └─6213 dhcpd -user dhcpd -group dhcpd -f -6 -pf /run/dhcp-server/dhcpd6.pid -cf /etc/dhcp/dhcpd6.conf ens33

Apr 14 23:05:22 ubuntu dhcpd[6213]: Sending Reply to fe80::59fa:47f4:8dac:5842 port 546
Apr 14 23:07:00 ubuntu dhcpd[6213]: Renew message from fe80::e68a:1d8c:ad22:424b port 546, transaction ID 0xD541F100
Apr 14 23:07:00 ubuntu dhcpd[6213]: Reply NA: address 2001:db8:27::170 to client with duid 00:01:00:01:2f:8c:9c:2c:00:0c:29:57:b3:d6 taId = 693613526 valid for 600 seco
Apr 14 23:07:00 ubuntu dhcpd[6213]: Sending Reply to fe80::e68a:1d8c:ad22:424b port 546
Apr 14 23:10:22 ubuntu dhcpd[6213]: Renew message from fe80::59fa:47f4:8dac:5842 port 546, transaction ID 0x18442900
Apr 14 23:10:22 ubuntu dhcpd[6213]: Reply NA: address 2001:db8:27::1b8 to client with duid 00:01:00:01:2f:8c:af:ad:00:0c:29:3e:26:48 taId = 691938888 valid for 600 seco
Apr 14 23:10:22 ubuntu dhcpd[6213]: Sending Reply to fe80::59fa:47f4:8dac:5842 port 546
Apr 14 23:12:00 ubuntu dhcpd[6213]: Renew message from fe80::e68a:1d8c:ad22:424b port 546, transaction ID 0xB827F000
Apr 14 23:12:00 ubuntu dhcpd[6213]: Reply NA: address 2001:db8:27::170 to client with duid 00:01:00:01:2f:8c:9c:2c:00:0c:29:57:b3:d6 taId = 693613526 valid for 600 seco
Apr 14 23:12:00 ubuntu dhcpd[6213]: Sending Reply to fe80::e68a:1d8c:ad22:424b port 546
lines 1-19/19 (END)
```

Fig: isc-dhcp-server status active

```

group1@ubuntu: ~
File Edit View Search Terminal Help
GNU nano 2.9.3 /etc/dhcp/dhcpd.conf

subnet 192.168.27.0 netmask 255.255.255.0 {
    range 192.168.27.100 192.168.27.200;
    option routers 192.168.27.2;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.27.255;
    option domain-name-servers 8.8.8.8, 8.8.4.4;

    pool {
        range 192.168.27.11 192.168.27.99;
        deny all clients;
    }
}

#use-host-decl-names on;

host Client1 {
    hardware ethernet 00:0c:29:57:b3:d6;
    option host-name "Client_1";
}

host Client2 {
    hardware ethernet 00:0c:29:3e:26:48;
    option host-name "Client_2";
}

host Client1-reservation {
    hardware ethernet 00:0c:29:57:b3:d6;
    fixed-address 192.168.27.133;
}

host Client2-reservation {
    hardware ethernet 00:0c:29:3e:26:48;
    fixed-address 192.168.27.135;
}

```

Fig: DHCP Configuration (v4)

```

group1@ubuntu: ~
File Edit View Search Terminal Help
GNU nano 2.9.3 /etc/dhcp/dhcpd6.conf

#subnet6 3ffe:501:ffff:101::/64 {
#    range6 3ffe:501:ffff:101::10 3ffe:501:ffff:101::11;
#
#    # Override of the global definitions,
#    # works only when a resource (address or prefix) is assigned
#    option dhcp6.name-servers 3ffe:501:ffff:101:200:ff:fe00:3f3e;
#}

# A third subnet behind a relay agent chain
#subnet6 3ffe:501:ffff:102::/64 {
#    range6 3ffe:501:ffff:102::10 3ffe:501:ffff:102::11;
#}

subnet6 2001:db8:27::/64 {
    range6 2001:db8:27::100 2001:db8:27::200;
    option dhcp6.name-servers 2001:4860:4860::8888, 2001:4860:4860::8844, 2001:db8:27::130, 2001:db8:27::131;
    option dhcp6.domain-search "group1-startup.local";
}

host Client1-v6 {
    host-identifier option dhcp6.client-id "\326\263W)\000\001\000\001\214\234,\000\014)\263\326";
    fixed-address6 2001:db8:27::200;
}

host Client2-v6 {
    host-identifier option dhcp6.client-id "H8>)\000\001\000\001\214\257\255\000\014)>8H";
    fixed-address6 2001:db8:27::139;
}

```

Fig: DHCP Configuration (v6)

```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
GNU nano 2.9.3 /etc/default/isc-dhcp-server  
  
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)  
  
# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).  
DHCPDv4_CONF=/etc/dhcp/dhcpd.conf  
DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf  
  
# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).  
#DHCPDv4_PID=/var/run/dhcpd.pid  
#DHCPDv6_PID=/var/run/dhcpd6.pid  
  
# Additional options to start dhcpd with.  
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead  
#OPTIONS=""  
  
# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?  
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".  
INTERFACESv4="ens33"  
INTERFACESv6="ens33"
```

Fig: IPv4 and IPv6 interface configuration

```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
group1@ubuntu:~$ sudo cat /var/lib/dhcp/dhcpd.leases  
# The format of this file is documented in the dhcpd.leases(5) manual page.  
# This lease file was written by isc-dhcp-4.3.5  
  
# authoring-byte-order entry is generated, DO NOT DELETE  
authoring-byte-order little-endian;  
  
lease 192.168.27.101 {  
  starts 5 2025/04/11 05:42:11;  
  ends 5 2025/04/11 05:52:11;  
  tstp 5 2025/04/11 05:52:11;  
  cltt 5 2025/04/11 05:42:11;  
  binding state free;  
  hardware ethernet 00:0c:29:3e:26:48;  
}  
lease 192.168.27.147 {  
  starts 6 2025/04/12 01:24:57;  
  ends 6 2025/04/12 01:34:57;  
  tstp 6 2025/04/12 01:34:57;  
  cltt 6 2025/04/12 01:24:57;  
  binding state free;  
  hardware ethernet 00:0c:29:42:a0:06;  
}  
lease 192.168.27.135 {  
  starts 6 2025/04/12 01:55:00;  
  ends 6 2025/04/12 02:05:00;  
  tstp 6 2025/04/12 02:05:00;  
  cltt 6 2025/04/12 01:55:00;  
  binding state free;  
  hardware ethernet 00:0c:29:3e:26:48;  
}  
lease 192.168.27.133 {  
  starts 6 2025/04/12 03:47:35;  
  ends 6 2025/04/12 03:57:35;  
  tstp 6 2025/04/12 03:57:35;  
  cltt 6 2025/04/12 03:47:35;  
  binding state free;  
  hardware ethernet 00:0c:29:57:b3:d6;  
}  
lease 192.168.27.138 {  
  starts 6 2025/04/12 07:29:14;  
  ends 6 2025/04/12 07:39:14;  
  tstp 6 2025/04/12 07:39:14;  
  cltt 6 2025/04/12 07:29:14;  
  binding state free;  
  hardware ethernet 00:0c:29:22:70:48;  
}
```

Fig: IPv4 Leases

```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
group1@ubuntu:~$ sudo cat /var/lib/dhcp/dhcpd6.leases  
# The format of this file is documented in the dhcpd.leases(5) manual page.  
# This lease file was written by isc-dhcp-4.3.5  
  
# authoring-byte-order entry is generated, DO NOT DELETE  
authoring-byte-order little-endian;  
  
server-duid "\000\001\000\001/\214\257\005\000\014)B\240\006";  
  
ia-na "H&>)\000\001\000\001/\214\257\255\000\014)>&H" {  
  cltt 2 2025/04/15 05:22:19;  
  iaaddr 2001:db8:27::1b8 {  
    binding state active;  
    preferred-life 375;  
    max-life 600;  
    ends 2 2025/04/15 05:32:19;  
  }  
}  
  
ia-na "\326\263W)\000\001\000\001/\214\234,\000\014)W\263\326" {  
  cltt 2 2025/04/15 05:23:57;  
  iaaddr 2001:db8:27::170 {  
    binding state active;  
    preferred-life 375;  
    max-life 600;  
    ends 2 2025/04/15 05:33:57;  
  }  
}  
  
ia-na "H&>)\000\001\000\001/\214\257\255\000\014)>&H" {  
  cltt 2 2025/04/15 05:22:19;  
  iaaddr 2001:db8:27::1b8 {  
    binding state expired;  
    preferred-life 375;  
    max-life 600;  
    ends 2 2025/04/15 05:32:19;  
  }  
}  
  
ia-na "\326\263W)\000\001\000\001/\214\234,\000\014)W\263\326" {  
  cltt 2 2025/04/15 05:23:57;  
  iaaddr 2001:db8:27::170 {  
    binding state expired;  
    preferred-life 375;  
    max-life 600;  
    ends 2 2025/04/15 05:33:57;  
  }  
}
```

Fig: IPv6 Leases

5. WEB SERVER IMPLEMENTATION

5.1. INTRODUCTION TO WEB SERVER

A web server is a software or hardware system that serves web content, such as websites, web applications, or files, to clients (typically web browsers) over the Internet or a local network. It handles incoming requests using the HTTP (Hypertext Transfer Protocol) or HTTPS protocol, processes them, and delivers the appropriate response, often in the form of HTML pages, images, or other resources. Common web server software includes Apache, Nginx, and Microsoft IIS, which are designed to ensure fast, secure, and efficient delivery of web content. Web servers play a critical role in enabling users to access and interact with online services and applications seamlessly.

5.2. BEHAVIOR OF WEB SERVER

Listening for Requests:

- The web server continuously listens on a specific port (usually port 80 for HTTP and 443 for HTTPS) for incoming client requests, typically from web browsers.

Receiving an HTTP Request:

- When a user enters a URL or clicks a link, the browser sends an HTTP request to the web server for the corresponding resource (like an HTML page, image, or script).

Processing the Request:

- The server processes the request by locating the requested file or executing server-side scripts (like PHP, Python, or JavaScript).

Generating a Response:

- The web server prepares an HTTP response, which includes a status code (like 200 OK, 404 Not Found) and the requested content.

Sending the Response:

- The server sends the HTTP response back to the client (browser), which then renders the content for the user to view and interact with.

Logging and Monitoring:

- Most web servers maintain logs of requests and responses for monitoring, debugging, analytics, and security auditing.

5.3. COMMANDS USED

Step Description	Commands
Install Apache	<code>sudo apt update</code> <code>sudo apt install apache2</code>
Configure Network	<code>sudo nano /etc/netplan/01-netcfg.yaml</code> <code>sudo netplan apply</code>
Check Apache Status	<code>sudo systemctl status apache2</code>
Start and Enable Apache	<code>sudo systemctl start apache2</code> <code>sudo systemctl enable apache2</code>
Edit Web Page	<code>sudo nano /var/www/html/index.html</code>
Set Permissions	<code>sudo chmod 644 /var/www/html/index.html</code>
Test Locally	<code>curl http://localhost</code>

5.4. TEST RESULTS

```
group1@ubuntu: /etc/netplan
File Edit View Search Terminal Help
GNU nano 2.9.3                                01-network-manager-all.yaml

# Let NetworkManager manage all devices on this system
#network:
# version: 2
# renderer: NetworkManager
network:
  version: 2
  renderer: NetworkManager
  ethernet:
    ens33:
      dhcp4: no
      addresses:
        - 192.168.27.132/24
        - 2001:db8:27::132/64
      gateway4: 192.168.27.2
      gateway6: 2001:db8:27::2
      nameservers:
        addresses: [192.168.27.130, "2001:db8:27::130", 192.168.27.131, "2001:db8:27::131", 8.8.8.8, "2001:4860:4860::8888", 8.8.4.4, "2001:4860:4860::8844"]
```

Fig: Apache2 Web Server Network Configuration

```
group1@ubuntu: ~
File Edit View Search Terminal Help

group1@ubuntu:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)
   Drop-In: /lib/systemd/system/apache2.service.d
            └─apache2-systemd.conf
   Active: active (running) since Sun 2025-04-13 16:47:40 PDT; 1 day 6h ago
     Process: 19794 ExecStop=/usr/sbin/apachectl stop (code=exited, status=0/SUCCESS)
     Process: 23845 ExecReload=/usr/sbin/apachectl graceful (code=exited, status=0/SUCCESS)
     Process: 19799 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 19804 (apache2)
     Tasks: 55 (limit: 2281)
    CGroup: /system.slice/apache2.service
            └─19804 /usr/sbin/apache2 -k start
              23877 /usr/sbin/apache2 -k start
              23884 /usr/sbin/apache2 -k start

Apr 13 16:47:40 ubuntu systemd[1]: Starting The Apache HTTP Server...
Apr 13 16:47:40 ubuntu apachectl[19799]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName'
Apr 13 16:47:40 ubuntu systemd[1]: Started The Apache HTTP Server.
Apr 14 07:35:13 ubuntu systemd[1]: Reloading The Apache HTTP Server.
Apr 14 07:35:13 ubuntu apachectl[23845]: AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 127.0.1.1. Set the 'ServerName'
Apr 14 07:35:13 ubuntu systemd[1]: Reloaded The Apache HTTP Server.
lines 1-21/21 (END)
```

Fig: Apache Web Server Status

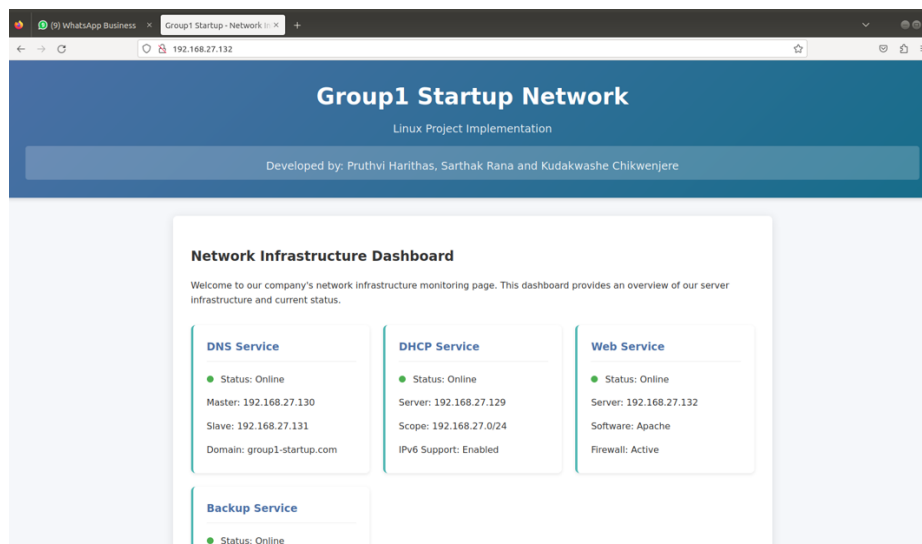


Fig: www.group1-startup.com

6. FIREWALL IMPLEMENTATION

6.1. INTRODUCTION TO FIREWALL

A firewall is a network security system, either hardware, software, or a combination of both that monitors and controls incoming and outgoing network traffic based on predefined security rules. Acting as a barrier between trusted internal networks and untrusted external networks (like the internet), firewalls help prevent unauthorized access, data breaches, and cyberattacks. They can filter traffic based on IP addresses, ports, protocols, and even specific content, providing a crucial layer of defense in modern network infrastructure. Firewalls are essential for protecting devices, servers, and data from malicious threats while allowing legitimate communication to flow smoothly.

6.2. BEHAVIOR OF FIREWALL

Packet Filtering:

- Inspects each packet of data that enters or leaves the network.
- Allows or blocks packets based on predefined rules (e.g., source/destination IP, port number, protocol).

Stateful Inspection:

- Keeps track of active connections and only allows packets that match a known active connection.
- Enhances security by analyzing the state and context of the traffic.

Proxy Functionality:

- Acts as an intermediary between internal and external networks.
- Requests from clients are forwarded to the internet, and responses are sent back through the proxy, hiding the internal network.

Logging and Monitoring:

- Maintains logs of all allowed and denied traffic.
- Helps administrators track suspicious activities and enforce security policies.

Rule-Based Decision Making:

- Makes real-time decisions using a set of rules or policies defined by the administrator.
- These rules determine what type of traffic is permitted or denied.

Protection Against Attacks:

- Identifies and blocks common threats such as unauthorized access, port scanning, and denial-of-service (DoS) attacks.

6.3. COMMANDS USED

Step Description	Commands
Install UFW	<code>sudo apt install ufw</code>
Check Firewall Status	<code>sudo ufw status</code>
Set Default Policies	<code>sudo ufw default deny incoming</code> <code>sudo ufw default allow outgoing</code>
Allow SSH Access	<code>sudo ufw allow ssh</code>
Allow Web Traffic	<code>sudo ufw allow 80/tcp</code> <code>sudo ufw allow 443/tcp</code>
Allow Internal Network	<code>sudo ufw allow from 192.168.27.0/24</code>
Enable Firewall	<code>sudo ufw enable</code>
Check Detailed Status	<code>sudo ufw status verbose</code>
Test Connections	<code>sudo iptables -L -v</code>

	<code>sudo ufw status</code>
Disable Firewall (for testing)	<code>sudo ufw disable</code>

6.4. TEST RESULTS

```

group1@ubuntu: ~
File Edit View Search Terminal Help
group1@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
88/tcp ALLOW Anywhere
443/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
88/tcp (v6) ALLOW Anywhere (v6)
443/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

53/tcp ALLOW OUT Anywhere
53/udp ALLOW OUT Anywhere
53/tcp (v6) ALLOW OUT Anywhere (v6)
53/udp (v6) ALLOW OUT Anywhere (v6)

group1@ubuntu:~$

```

Fig: Firewall Status

```

group1@ubuntu: ~
File Edit View Search Terminal Help
group1@ubuntu:~$ curl http://localhost
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>403 Forbidden</title>
</head><body>
<h1>Forbidden</h1>
<p>You don't have permission to access this resource.</p>
</body></html>
group1@ubuntu:~$ curl http://www.group1-startup.com
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>403 Forbidden</title>
</head><body>
<h1>Forbidden</h1>
<p>You don't have permission to access this resource.</p>
</body></html>
group1@ubuntu:~$ curl http://192.168.27.132
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>403 Forbidden</title>
</head><body>
<h1>Forbidden</h1>
<p>You don't have permission to access this resource.</p>
</body></html>
group1@ubuntu:~$

```

Fig: Firewall blocking the web page source code access

7. BACKUP SERVER

7.1. INTRODUCTION OF BACKUP SERVER

A backup server is a dedicated system used to store copies of data, applications, and system configurations from other computers or servers to ensure data safety and continuity in case of failures, loss, or disasters. It acts as a centralized repository where backup jobs are scheduled and managed, enabling quick recovery of critical information when needed. Backup servers can be configured to perform regular backups, either full, incremental, or differential, based on the organization's policies. They play a vital role in data protection strategies, helping minimize downtime and data loss while ensuring business continuity.

7.2. BEHAVIOR OF BACKUP SERVER

Scheduled Backups:

- Performs backups automatically at predefined intervals (daily, weekly, or hourly), ensuring that data is consistently saved without manual intervention.

Data Collection:

- Connects to client machines or servers over the network to collect and copy files, databases, system settings, and other important data.

Backup Types Handling:

Supports different types of backups:

- **Full Backup** – Copies all data.
- **Incremental Backup** – Copies only the data changed since the last backup.
- **Differential Backup** – Copies data changed since the last full backup.

Compression and Encryption:

- Often compresses backup data to save storage space and encrypts it to maintain confidentiality and security.

Storage Management:

- Organizes backups efficiently on local disks, external drives, network-attached storage (NAS), or cloud platforms.

Monitoring and Alerts:

- Continuously monitors backup tasks and notifies the admin in case of failures, storage issues, or missed schedules.

Data Restoration:

- Allows users or administrators to restore data fully or selectively in case of system failure, data corruption, or accidental deletion.

7.3. COMMANDS USED

Step Description	Commands
Configure Network	<code>sudo nano /etc/netplan/01-netcfg.yaml</code> <code>sudo netplan apply</code>
Install SSH Server	<code>sudo apt update</code> <code>sudo apt install openssh-server</code>
Start and Enable SSH	<code>sudo systemctl start ssh</code> <code>sudo systemctl enable ssh</code>
Allow SSH in Firewall	<code>sudo ufw allow ssh</code> <code>sudo ufw enable</code>

Create Backup Directories	<code>sudo mkdir -p /backups/{dns-master,dns-slave,dhcp,web}</code> <code>sudo chmod -R 777 /backups</code>
Create Backup Script	<code>sudo nano /opt/backup.sh</code>
Make Script Executable	<code>sudo chmod +x /opt/backup.sh</code>
Schedule Automated Backups	<code>sudo crontab -e</code>
Install Compression Tools	<code>sudo apt install zip</code>
Test Manual Backup	<code>sudo /opt/backup.sh</code>
Check Backup Status	<code>ls -l /backups/</code>
View Log	<code>cat /var/log/backups.log</code>

7.4. TEST RESULTS

```
group1@ubuntu: /etc/netplan
File Edit View Search Terminal Help
GNU nano 2.9.3                                01-network-manager-all.yaml

# Let NetworkManager manage all devices on this system
network:
# version: 2
# renderer: NetworkManager

network:
version: 2
renderer: NetworkManager
ethernets:
  ens33:
    dhcp4: no
    addresses:
      - 192.168.27.134/24
      - 2001:db8:27::134/64
    gateway4: 192.168.27.2
    gateway6: 2001:db8:27::2
    nameservers:
      addresses: [192.168.27.130, "2001:db8:27::130", 192.168.27.131, "2001:db8:27::131", 8.8.8.8, "2001:4860:4860::8888", 8.8.4.4, "2001:4860:4860::8844"]
```

Fig: Backup Server Network Configuration

```
group1@ubuntu: ~
File Edit View Search Terminal Help
zip is already the newest version (3.0-11build1).
The following packages were automatically installed and are no longer required:
  fonts-liberation2 fonts-opensymbol gir1.2-goa-1.0 gir1.2-gst-plugins-base-1.0 gir1.2-gstreamer-1.0 gir1.2-gudev-1.0 gir1.2-snapd-1 gir1.2-udisks-2.0
  grilo-plugins-0.3-base gstreamer1.0-gtk3 libboost-date-time1.65.1 libboost-filesystem1.65.1 libboost-locale1.65.1 libcdt-0.1-1
  libclucene-contribs1v5 libclucene-core1v5 libcmis-0.5-svs libcoland2 libdazzle-1.0-0 libe-book-0.1:1 libedataserverui-1.2-2 libboot0 libbepubgen-0.1-1 libbetonyek-0.1-1
  libevent-2.1-6 libexiv2-14 libfreerdp-client2-2 libfreerdp2-2 libgic2 libgee-0.8-2 libgexiv2-2 libgon-1.0-0 libgpgmepp0 libgpod-common libgpod4 liblangtag-common
  liblangtag1 liblirc-client0 liblua5.3-0 libmediaart-2.0-0 libmshpub-0.1-1 libodfgen-0.1-1 libqwing2v5 libraw16 librevenge-0.0-0 libsgutils2-2 libssh-4
  libsuitesparseconfig5 libvncclient1 libwinpr2-2 libxapian30 libxnlsec1-nss lp-solve media-player-info python3-nako python3-markupsafe syslinux syslinux-common
  syslinux-legacy usb-creator-common
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
adding: backups/dns-master/dns-master-2025-04-13.txt (stored 0%)
adding: backups/dns-master/dns-master-2025-04-14.txt (stored 0%)
adding: backups/dns-slave/dns-slave-2025-04-13.txt (stored 0%)
adding: backups/dns-slave/dns-slave-2025-04-14.txt (stored 0%)
adding: backups/dhcp/dhcp-2025-04-13.txt (stored 0%)
adding: backups/dhcp/dhcp-2025-04-14.txt (stored 0%)
adding: backups/web/web-2025-04-13.txt (stored 0%)
adding: backups/web/web-2025-04-14.txt (stored 0%)
Mon Apr 14 23:28:40 PDT 2025: Backup completed successfully. Files saved to /backups/
Backup completed successfully. Files saved to /backups/
Mon Apr 14 23:28:40 PDT 2025: Master backup ZIP created: /backups/all-servers-backup-2025-04-14.zip
/backups/dhcp/:
total 8
-rw-r--r-- 1 root root 33 Apr 13 10:05 dhcp-2025-04-13.txt
-rw-r--r-- 1 root root 33 Apr 14 23:28 dhcp-2025-04-14.txt

/backups/{dns-master,/}:
total 0

/backups/dns-master/:
total 16
-rw-r--r-- 1 root root 5051 Apr 13 08:52 dns-master-2025-04-13.tar.gz
-rw-r--r-- 1 root root 39 Apr 13 10:05 dns-master-2025-04-13.txt
-rw-r--r-- 1 root root 39 Apr 14 23:28 dns-master-2025-04-14.txt

/backups/dns-slave/:
total 16
-rw-r--r-- 1 root root 4429 Apr 13 00:42 dns-slave-2025-04-13.tar.gz
-rw-r--r-- 1 root root 38 Apr 13 10:05 dns-slave-2025-04-13.txt
-rw-r--r-- 1 root root 38 Apr 14 23:28 dns-slave-2025-04-14.txt

/backups/web/:
total 8
-rw-r--r-- 1 root root 32 Apr 13 10:05 web-2025-04-13.txt
-rw-r--r-- 1 root root 32 Apr 14 23:28 web-2025-04-14.txt
-rw-r--r-- 1 root root 972 Apr 13 10:05 /backups/all-servers-backup-2025-04-13.zip
-rw-r--r-- 1 root root 1922 Apr 14 23:28 /backups/all-servers-backup-2025-04-14.zip
group1@ubuntu:~$
```

Fig: Backup Logs

8. MAN-IN-THE-MIDDLE ATTACK (MITM)

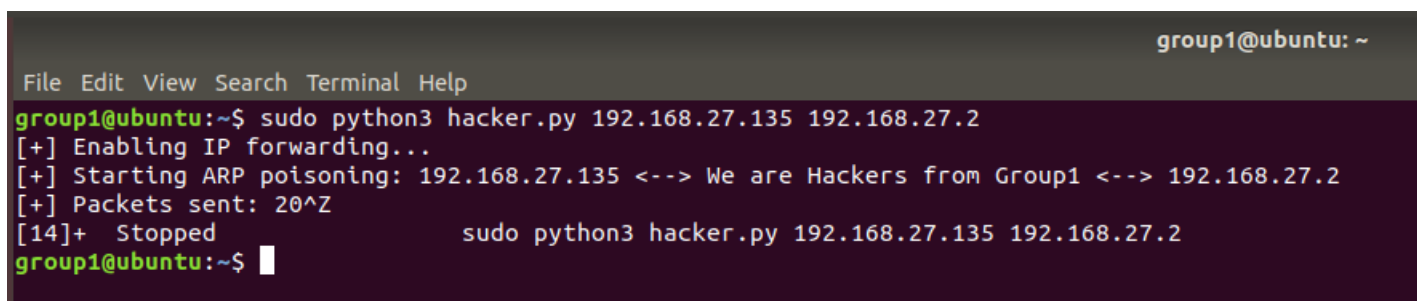
8.1. INTRODUCTION

A **Man-in-the-Middle (MITM) attack** is a type of cyberattack where an attacker secretly intercepts, alters, or injects malicious data into communication between two parties (e.g., a client and a server) without their knowledge. This can happen in any form of communication, such as HTTP, email, or instant messaging. The attacker can eavesdrop on sensitive information like passwords, credit card details, or private messages, often leading to identity theft, data breaches, or fraud. MITM attacks are commonly executed on insecure networks, such as public Wi-Fi, where the attacker can position themselves between the victim and the intended recipient.

8.2. COMMANDS USED

Step Description	Commands
Install Required Packages	<code>sudo apt update</code> <code>sudo apt install python3-pip tcpdump</code> <code>sudo pip3 install scapy</code>
Create Attack Script	<code>sudo nano mitm_attack.py</code>
Make Script Executable	<code>sudo chmod +x mitm_attack.py</code>
Execute Attack	<code>sudo python3 mitm_attack.py 192.168.27.135 192.168.27.2</code>
Capture Traffic	<code>sudo tcpdump -n host 192.168.27.135</code>
Check ARP Tables	<code>arp -a</code>
Enable IP Forwarding	<code>echo 1 > /proc/sys/net/ipv4/ip_forward</code>

8.3. TEST RESULTS



```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
group1@ubuntu:~$ sudo python3 hacker.py 192.168.27.135 192.168.27.2  
[+] Enabling IP forwarding...  
[+] Starting ARP poisoning: 192.168.27.135 <--> We are Hackers from Group1 <--> 192.168.27.2  
[+] Packets sent: 20^Z  
[14]+ Stopped sudo python3 hacker.py 192.168.27.135 192.168.27.2  
group1@ubuntu:~$
```

Fig: Man-in-the-Middle Attack

9. IPsec VPN TUNNELING

9.1. INTRODUCTION

IPsec (Internet Protocol Security) is a framework for securing IP communications by encrypting and authenticating all traffic at the IP layer. IPsec VPN tunneling uses IPsec protocols to create a secure "tunnel" between two endpoints, such as between a client and a VPN server, or between two branch offices. This tunnel ensures that the data transmitted between the endpoints is confidential, intact, and authenticated. It operates by using encryption techniques like ESP (Encapsulating Security Payload) and AH (Authentication Header) to protect data from eavesdropping and tampering. IPsec VPNs are widely used for secure remote access to corporate networks or for inter-site communication over untrusted networks like the internet.

9.2. COMMANDS USED

IPsec VPN Tunnel

Step Description	Commands
Install Strongswan	<pre>sudo apt update sudo apt install strongswan strongswan-pki libcharon-extra-plugins</pre>
Configure IPsec	<pre>sudo nano /etc/ipsec.conf</pre>
Set Up Pre-shared Key	<pre>sudo nano /etc/ipsec.secrets</pre>
Enable IP Forwarding	<pre>sudo nano /etc/sysctl.conf sudo sysctl -p</pre>
Restart IPsec Service	<pre>sudo systemctl restart strongswan</pre>
Check VPN Status	<pre>sudo ipsec status</pre>
Test Connection	<pre>ping 192.168.27.135</pre>
View Encrypted Traffic	<pre>sudo tcpdump -n esp</pre>

NFS for File Sharing

Step Description	Commands
Install NFS Server	<pre>sudo apt update sudo apt install nfs-kernel-server</pre>
Create Share Directory	<pre>sudo mkdir -p /var/nfs/general sudo chown nobody:nogroup /var/nfs/general</pre>
Configure Exports	<pre>sudo nano /etc/exports</pre>
Export Filesystem	<pre>sudo exportfs -a sudo systemctl restart nfs-kernel-server</pre>
Install NFS Client	<pre>sudo apt install nfs-common</pre>
Create Mount Point	<pre>sudo mkdir -p /mnt/nfs/general</pre>
Mount NFS Share	<pre>sudo mount 192.168.27.130:/var/nfs/general /mnt/nfs/general</pre>
Set Up Auto-mount	<pre>sudo nano /etc/fstab</pre>
Test File Access	<pre>touch /mnt/nfs/general/test_file ls -l /mnt/nfs/general/</pre>

9.3. Test Results:

```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
group1@ubuntu:~$ sudo ipsec status  
Security Associations (1 up, 0 connecting):  
client1-to-client2[16]: ESTABLISHED 22 minutes ago, 192.168.27.133[client1]...192.168.27.135[client2]  
client1-to-client2{16}:  INSTALLED, TUNNEL, reqid 16, ESP SPIs: c8be05fe_i c832e0df_o  
client1-to-client2{16}:  192.168.27.133/32 === 192.168.27.135/32  
group1@ubuntu:~$
```

Fig: IPSec Status in Client 1

```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
group1@ubuntu:~$ sudo ipsec status  
Security Associations (1 up, 0 connecting):  
client2-to-client1[17]: ESTABLISHED 21 minutes ago, 192.168.27.135[client2]...192.168.27.133[client1]  
client2-to-client1{16}:  INSTALLED, TUNNEL, reqid 16, ESP SPIs: c832e0df_i c8be05fe_o  
client2-to-client1{16}:  192.168.27.135/32 === 192.168.27.133/32  
group1@ubuntu:~$
```

Fig: IPSec Status in Client 2

```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
group1@ubuntu:~$ sudo tcpdump -n esp  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on ens33, link-type EN10MB (Ethernet), capture size 262144 bytes  
23:57:20.980127 IP 192.168.27.133 > 192.168.27.135: ESP(spi=0xc832e0df,seq=0x7d), length 136  
23:57:20.980336 IP 192.168.27.135 > 192.168.27.133: ESP(spi=0xc8be05fe,seq=0x97), length 136  
23:57:21.983329 IP 192.168.27.133 > 192.168.27.135: ESP(spi=0xc832e0df,seq=0x7e), length 136  
23:57:21.983450 IP 192.168.27.135 > 192.168.27.133: ESP(spi=0xc8be05fe,seq=0x98), length 136  
23:57:22.986180 IP 192.168.27.133 > 192.168.27.135: ESP(spi=0xc832e0df,seq=0x7f), length 136  
23:57:22.986350 IP 192.168.27.135 > 192.168.27.133: ESP(spi=0xc8be05fe,seq=0x99), length 136  
23:57:23.988142 IP 192.168.27.133 > 192.168.27.135: ESP(spi=0xc832e0df,seq=0x80), length 136  
23:57:23.988579 IP 192.168.27.135 > 192.168.27.133: ESP(spi=0xc8be05fe,seq=0x9a), length 136  
23:57:24.990524 IP 192.168.27.133 > 192.168.27.135: ESP(spi=0xc832e0df,seq=0x81), length 136  
23:57:24.990710 IP 192.168.27.135 > 192.168.27.133: ESP(spi=0xc8be05fe,seq=0x9b), length 136  
23:57:25.993941 IP 192.168.27.133 > 192.168.27.135: ESP(spi=0xc832e0df,seq=0x82), length 136  
23:57:25.994069 IP 192.168.27.135 > 192.168.27.133: ESP(spi=0xc8be05fe,seq=0x9c), length 136
```

Fig: ESP Logs indicate encrypted message transfer through the tunnel

```
group1@ubuntu: ~  
File Edit View Search Terminal Help  
group1@ubuntu:~$ ls -l /mnt/nfs/shared/  
total 4  
-rw-rw-r-- 1 group1 group1 40 Apr 14 23:48 server_test.txt  
-rw-r--r-- 1 nobody nogroup 0 Apr 14 23:44 test_file  
group1@ubuntu:~$ cat /mnt/nfs/shared/server_test.txt  
This is a test file from the NFS server  
group1@ubuntu:~$
```

Fig: NFS for file sharing (File shared from Client 1 to Client 2)

10. Linux System Topics

10.1. File System of Linux

The Linux file system follows a hierarchical structure with a single root directory ("/") from which all other directories branch out. It adopts the "everything is a file" philosophy, where system resources, devices, and even processes are represented as files. Key directories include /bin for binaries, /etc for configuration files, /home for user files, /var for variable data, and /proc which provides a view into kernel processes. This unified approach offers flexibility and consistency for system administration.

tar, gunzip

The tar (tape archive) utility in Linux is essential for bundling multiple files into a single archive without compression. It's commonly combined with compression tools like gzip to create compressed archives. For example, "tar -czvf backup.tar.gz /var/www/" creates a compressed archive of web files. The gunzip utility specifically decompresses files compressed with gzip. These utilities are foundational for backup operations, software distribution, and data transfer in Linux environments.

10.2. Memory

Linux manages memory through a sophisticated virtual memory system that abstracts physical RAM and extends it using swap space. The kernel dynamically allocates memory to processes, caches frequently accessed data, and implements demand paging to load only needed memory segments. Tools like "free" and "top" allow administrators to monitor memory usage, while the /proc/meminfo file provides detailed memory statistics. Linux's efficient memory management enables it to run effectively on systems with varying hardware specifications.

10.3. Processes

In Linux, processes represent running programs, each with a unique PID (Process ID). The kernel schedules these processes using sophisticated algorithms that balance system responsiveness and resource utilization. Processes operate within their own memory space for stability and security. The process hierarchy is structured as a tree with init (PID 1) as the root. Administrators can manage processes using commands like ps, top, and kill, while the /proc filesystem provides programmatic access to process information.

11. Future Scope

Future developments in Linux networking could include more integrated software-defined networking (SDN) capabilities, enhanced security through automated threat detection, improved IPv6 implementation, and more seamless container networking. Automation through technologies like Ansible and Terraform will likely become more sophisticated, while emerging AI-based network management tools may offer predictive maintenance and anomaly detection. Additionally, integration with cloud-native technologies will continue to evolve, enabling hybrid deployments across on-premises and cloud environments.

12. References

- Nemeth, E., Snyder, G., Hein, T. R., Whaley, B., & Mackin, D. (2017). UNIX and Linux System Administration Handbook (5th ed.). Addison-Wesley Professional.
- Hunt, C. (2021). TCP/IP Network Administration (4th ed.). O'Reilly Media.
- Barrett, D. J., Silverman, R. E., & Byrnes, R. G. (2017). Linux Security Cookbook (2nd ed.). O'Reilly Media.
- Liu, C. (2020). DNS and BIND (6th ed.). O'Reilly Media.
- Ubuntu Server Documentation. <https://ubuntu.com/server/docs>
- Red Hat Enterprise Linux Documentation. https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/
- The Linux Documentation Project. <https://tldp.org/>
- Internet Systems Consortium (ISC) BIND Documentation. <https://www.isc.org/bind/>
- Internet Systems Consortium (ISC) DHCP Documentation. <https://www.isc.org/dhcp/>
- Apache HTTP Server Documentation. <https://httpd.apache.org/docs/>
- Strongswan VPN Documentation. <https://docs.strongswan.org/>
- DigitalOcean Community Tutorials. <https://www.digitalocean.com/community/tutorials>
- Linux Journal. <https://www.linuxjournal.com/>
- IETF RFC Documents. <https://www.ietf.org/standards/rfcs/>
- The Tcpdump/Libpcap Documentation. <https://www.tcpdump.org/>
- YouTube. (2020). DNS Server Configuration in Linux | BIND DNS Server Configuration. <https://www.youtube.com/watch?v=4yKN9r48vo0>
- YouTube. (2017). How to Configure DHCP Server in Ubuntu Linux. <https://www.youtube.com/watch?v=9DtQRU7XHqw>
- YouTube. (2023). Setup Linux Web Server in Ubuntu | Apache2. https://youtu.be/XtRXm4FFK7Q?si=TVJOEiRNWtxL_B79